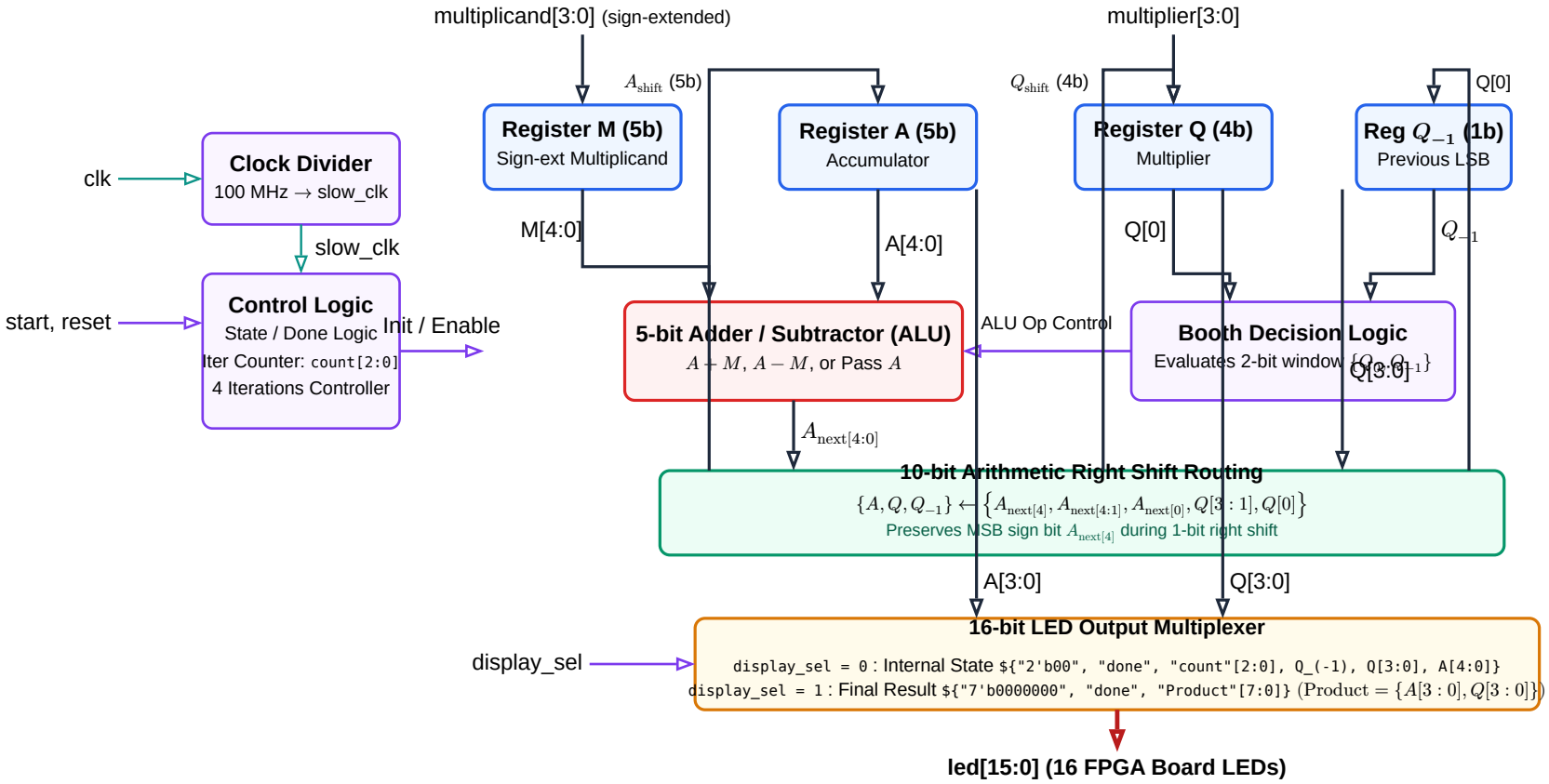
Datapath & Control Architecture: 4-bit Sequential Booth Multiplier

Iterative Signed 2's-Complement Hardware Architecture (IIT Kharagpur COA Lab)



Booth ALU Operation Table

$\{Q_0, Q_{-1}\}$	ALU Action	Operation Details
00	No Operation	Pass Accumulator ($A_{next} = A$)
01	$A \leftarrow A + M$	Add multiplicand to Accumulator
10	$A \leftarrow A - M$	Subtract multiplicand ($A + \widetilde{M} + 1$)
11	No Operation	Pass Accumulator ($A_{next} = A$)

LED Display Selection (`display_sel`)

`display_sel`	Mode	LED Vector Output (`led[15:0]`)
0	Internal State	`{ 2'b00, done, count[2:0], Q_-1, Q[3:0], A[4:0] }`
1	Final Result	`{ 7'b0000000, done, product[7:0] }` (product $= \{A[3:0], Q[3:0]\}$)